UNIT 7 · TOPIC 7

Carbonization and Densification — Briquetting and Pelletization

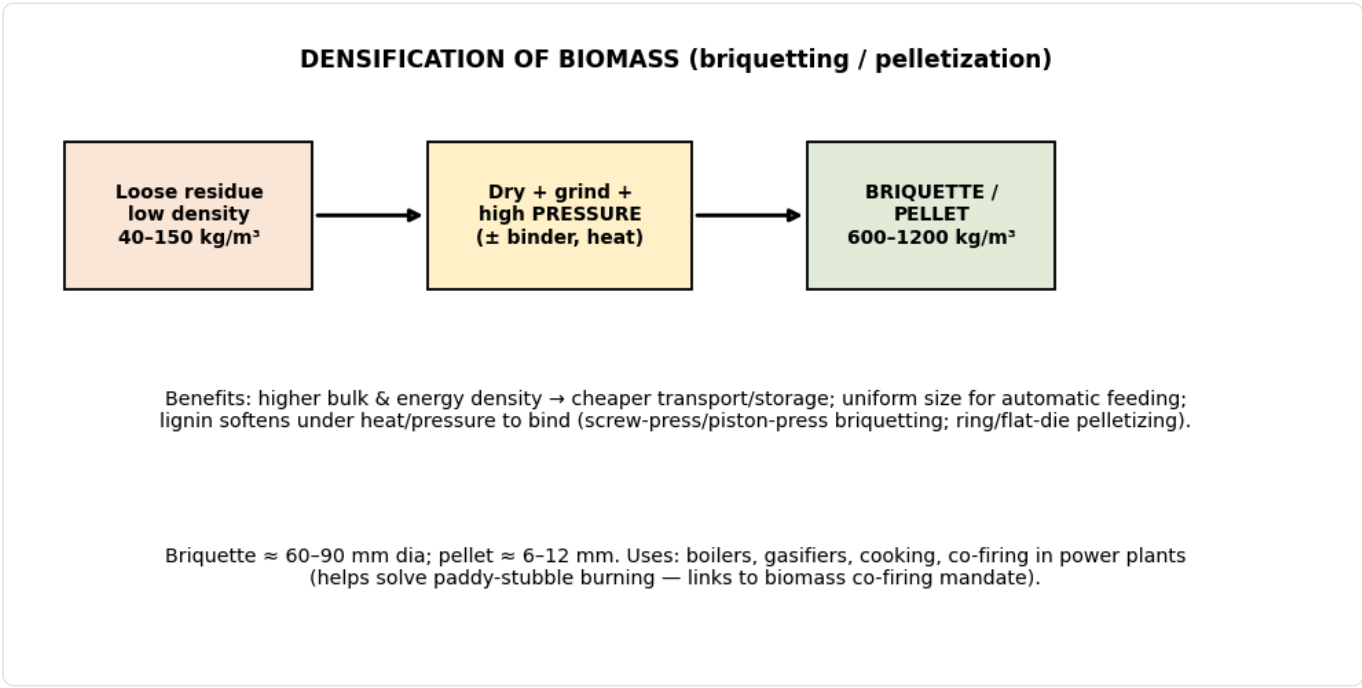
Syllabus topic: Chemical conversion processes: carbonization, briquetting, pelletization and densification of biomass

Carbonization

- **CARBONIZATION** = slow pyrolysis optimised to make **CHARCOAL** — a high-carbon, smokeless, high-CV solid fuel; the traditional kiln process.
- **Biochar** (a pyrolysis char) is also used as a soil amendment and carbon sink.

Why densify biomass

Loose agro-residues have very low bulk density (**40–150 kg/m³**), making them costly to collect, transport, store and feed.



Densification of biomass — briquetting and pelletization

The process

1. Dry the feedstock to ~**8–12%** moisture.
2. Grind it and control the particle size.
3. Compress it under HIGH PRESSURE — often with heat, sometimes with a binder.
4. The natural **LIGNIN** softens under heat and pressure and binds the particles on cooling (binderless screw-press or piston-press briquetting; ring-die or flat-die pelletizing).

Product	Size	Density
BRIQUETTES	~60–90 mm	600–1200 kg/m³
PELLETS	~6–12 mm	600–1200 kg/m³

Benefits

- Much higher bulk and energy density → cheaper logistics and storage.
- Uniform size for automatic stoker/gasifier feeding.
- Cleaner, higher-efficiency combustion.
- A productive use for surplus straw that reduces STUBBLE BURNING — biomass pellets are co-fired in thermal power plants under India's co-firing policy.

Practice MCQs

Q95. Slow pyrolysis / carbonization maximises:

(a) Bio-oil (b) Charcoal (char)
(c) Gas (d) Steam

Q96. Charcoal is produced by:

(a) Combustion (b) Carbonization (slow pyrolysis)
(c) Gasification (d) Fermentation

Q97. Biochar, a pyrolysis product, is also used as a:

(a) Fuel only (b) Soil amendment / carbon sink
(c) Fertiliser N source (d) Pesticide

Q98. Loose agro-residues have a bulk density of roughly:

(a) 40–150 kg/m³ (b) 600–1200 kg/m³
(c) 2000 kg/m³ (d) 5000 kg/m³

Q99. After densification, briquettes/pellets reach a density of about:

(a) 40–150 (b) 300
(c) 600–1200 (d) 3000 kg/m³

Q100. In binderless briquetting, the natural binder that softens under heat/pressure is:

(a) Cellulose (b) Lignin
(c) Starch (d) Silica

Q101. Typical pellet diameter is about:

(a) 1 mm (b) 6–12 mm
(c) 60–90 mm (d) 150 mm

Q102. A key benefit of densification is:

(a) Higher moisture (b) Higher bulk/energy density → cheaper transport & storage
(c) Lower CV (d) Larger size

Q103. Feedstock moisture before briquetting is typically reduced to about:

(a) 8–12% (b) 30%
(c) 50% (d) 70%

Q104. Biomass pellets help reduce which problem in North India?

(a) Soil salinity (b) Paddy-stubble burning (via co-firing)
(c) Groundwater (d) Wind erosion

Answers

Q95 — (b) Charcoal (char) — *Slow pyrolysis/carbonization → charcoal.*
Q96 — (b) Carbonization (slow pyrolysis)
Q97 — (b) Soil amendment / carbon sink

Q98 — (a) 40–150 kg/m³ — *Densification raises 40–150 → 600–1200 kg/m³.*
Q99 — (c) 600–1200
Q100 — (b) Lignin — *Lignin binds.*
Q101 — (b) 6–12 mm — *Pellets 6–12 mm; briquettes 60–90 mm.*
Q102 — (b) Higher bulk/energy density → cheaper transport & storage
Q103 — (a) 8–12% — *Dry to ~8–12%.*
Q104 — (b) Paddy-stubble burning (via co-firing)

Sources: Rai — Non-Conventional Energy Sources; Sukhatme; Duffie & Beckman; MNRE Bio-Energy & Wind programmes and PIB releases (as given in the source notes, July 2026); Betz (1919) actuator-disc theory; NPTEL renewable-energy & biomass lectures.